

Contoso Infrastructure — Harbour Bridge Maintenance Project Close-Out Report

Project Summary

Field	Details
Project Name	Harbour Bridge Deck Replacement & Strengthening
Client	Maritime Infrastructure Authority
Contract Type	Construct Only
Contract Value	\$128M (original), \$141M (final)
Duration	24 months (Mar 2024 – Feb 2026)
Location	Newcastle, NSW (marine environment)
Scope	850m bridge deck replacement, structural strengthening, pile remediation

Financial Performance

Budget vs Actual

Cost Category	Budget (\$M)	Actual (\$M)	Variance
Marine works & cofferdams	22.5	28.1	+24.9%
Concrete (deck & piles)	31.2	34.8	+11.5%
Structural steel	18.6	19.2	+3.2%
Pile remediation	14.8	16.4	+10.8%
Temporary works	12.0	14.5	+20.8%
Traffic management	8.4	8.1	-3.6%
Preliminaries	15.5	15.9	+2.6%
Contingency (used)	5.0	4.0	—
TOTAL	128.0	141.0	+10.2%

Key Unit Rates Achieved

Item	Tendered Rate	Actual Rate	Variance
Concrete 50MPa (marine grade)	\$395.00/m ³	\$425.00/m ³	+7.6%
Reinforcement (stainless, marine)	\$5,200/t	\$5,650/t	+8.7%

Item	Tendered Rate	Actual Rate	Variance
Bored piles (1200mm, marine)	\$1,100/lin m	\$1,280/lin m	+16.4%
Formwork (over water)	\$165.00/m ²	\$185.00/m ²	+12.1%
Cathodic protection	\$280/m ²	\$295/m ²	+5.4%

Concrete Supply Issues

Problem Statement

Marine-grade concrete (50MPa with supplementary cementite materials) experienced significant supply disruption during months 8-14 of the project:

- 1. **Supplier capacity** — Primary supplier (Boral) allocated to competing project; reduced weekly deliveries from 400m³ to 250m³
- 2. **Quality rejections** — 12% of delivered batches rejected due to chloride content exceeding AS 1379 limits for marine exposure
- 3. **Weather windows** — Marine concrete pours limited to weather windows (wave height < 0.5m); missed windows due to supply delays cascaded through program

Cost Impact

Issue	Additional Cost
Supplier premium (secondary supplier)	\$1.2M
Standby costs during supply gaps	\$0.8M
Quality testing (increased frequency)	\$0.3M
Program extension (2 weeks)	\$1.5M
Total concrete-related overrun	\$3.8M

Mitigation Actions Taken

- 1. Engaged secondary concrete supplier (Holcim) at 8% premium
- 2. Implemented on-site chloride testing (results in 2 hours vs 7-day lab test)
- 3. Negotiated priority allocation with Boral for months 14-20
- 4. Re-sequenced pours to maximize available weather windows

Lessons Learned

What Went Well

- 1. **Cathodic protection** — new anode system installed 15% under budget due to competitive supplier market
- 2. **Traffic management** — counter-flow arrangement maintained 2 lanes in each direction throughout
- 3. **Pile inspection** — underwater drone survey reduced diver time by 60%

What Could Improve

1. **Concrete supply** — single-supplier dependency for marine-grade concrete; always contract 2 suppliers from day 1
2. **Tidal windows** — underestimated impact of tidal cycles on cofferdam works; add 30% float to marine activity durations
3. **Temporary works** — barge-mounted crane costs escalated due to 8-week availability gap; book marine plant 6 months ahead
4. **Cost variance** — 10.2% overrun primarily driven by marine access difficulty and concrete supply; marine projects need minimum 12% contingency

Recommendations for Future Projects

- Dual-supply contracts mandatory for marine-grade concrete
- Include on-site rapid chloride testing in all marine concrete specifications
- Book marine plant (barges, jack-ups, crane barges) minimum 6 months prior
- Marine project contingency: minimum 12% (vs standard 5-8% for land-based)
- Engage specialist marine construction manager from tender stage

This document is for workshop demonstration purposes only. All data is fictional.